



Tools of the Trade: Linux and SQL

Module 1

Operating system (OS): the interface between computer hardware and the user.

Computer communicates in binary the OS provides an interface to bridge communication gap between user and computer.

User → Application → OS → Hardware

Hardware → OS → Application → User

OS handles resource and memory management

To ensure the limited capacity of the computer system is used where it is needed most.

Task manager lists all the task being performed along with memory and cpu usage.

User interface: A program that allows the user to control the functions of the operating systems.

Graphical user interface (GUI): a user interface that uses icons on the screen to manage different tasks on the computer.

Start menu

Task bar

Desktop with icons and shortcuts

Comand line interface (CLI): a text-based user interface that uses commands to interact with the computer.

Flexible

Powerful

Module 2

Linux: An open source operating system

Linux architecture

User: Person interacting with a computer

Applications: A program that performs a specific tasks

Shell: The command line interpreter

Filesystem Hierarchy standard (FHS): The component of linux OS that organizes data

Kernel: The component of the linux OS that manages processes and memory.

- Communicates with hardware to execute commands sent by the shell

- Uses drivers to enable to execute tasks

- Allocate memory more efficiently

Hardware: Physical components of a computer

Linux is a multi user system, meaning more than one user can use the systems resources at the same time.

Linux distributions: Different versions of linux

distros

Parent distributions:

- Red hat (CentOS)

- Slackware (SUSE)

- Debian (Ubuntu and KALI)

Penetration test: simulated attack that helps identify vulnerabilities in systems, networks, websites, applications, and processes.

Metasploit

burpsuite

john the ripper

Digital forensics: the practice of collecting and analyzing data to determine what has happened after an attack.

tcpdump

wireshark

autopsy

Package: a piece of software that can be combined with other packages to form an application.

Packages contain the files necessary for an application to be installed. these files include dependencies, which are supplemental files used to run an application.

Package manager is a tool that helps users install, manage and remove packages or applications.

Linux has multiple package manager.

Important to use the most recent version of a package when possible. The most recent version has the most up to date bug fixes and security patches.

command: an instruction telling the computer to do something

Shell: command line interpreter

Bourne-Again Shell (BASH)

C Shell (csh)

Korn Shell (ksh)

Enhanced C shell (tcsh)

Z Shell (zsh)

Standard input: information received by the OS via command line

echo: a linux command that outputs a specified string of text

string data: data consisting of an ordered sequence of characters

Standard output: information returned by the OS through the shell

Standard error: error messages returned by the OS through the shell

Module 3

Bash: the default shell in most linux distros

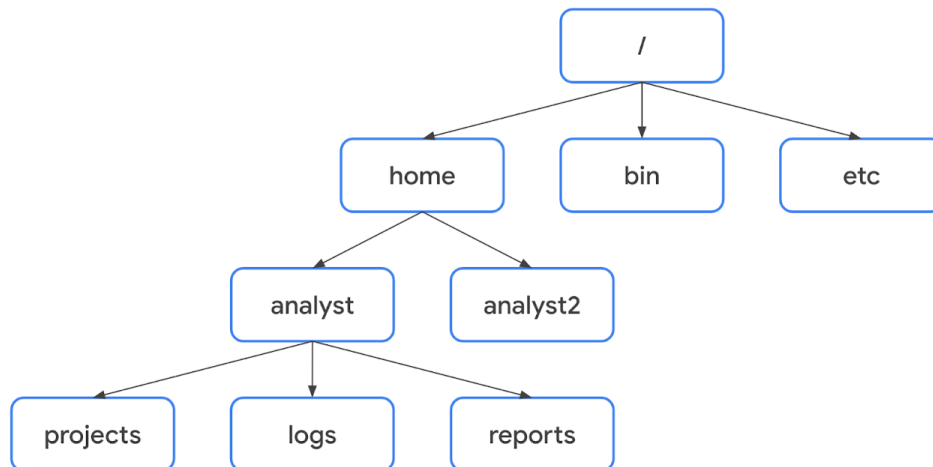
Argument (Linux): Specific information needed by a command

Filesystem hierarchy standard (FHS): The component of linux OS that organizes data.

Root Directory (/): the highest level directory in linux

Sub directory branch out from the root directory

```
graph TD; root_dir[root dir] --> home[home]; home --> analyst[analyst]
```



/home : each user in the system gets on dir

/bin :dir stands for "binary" and contains binary files and other executables. Executables are files that contain a series of commands a computer needs to follow to run programs and perform other functions.

/etc : dir stores the systems config files

/tmp : dir stores many temp files, tmp dir is commonly used by attackers because anyone in the system can modify data in these files.

/mnt : dir stands for "mount" and stores media, such as USB drives and hard drives.

Absolute file path is the full path that starts from the root

Relative file path is a shorter file path that start from the current directory

whoami : returns the current user

pwd : prints the working directory

ls : displays the names of files and dirs in the current working dir

cd : navigates between dirs

grep : searches a specified file and returns all lines in th file containing a specified string

| (piping) : send the standard output of one command as standard input to another command for further processing.

find : searches for directories and files that meet specified criteria.

find /home/analyst/projects -name "*log*" : case sensitive

find /home/analyst/projects -iname "*log*" : not case sensitive

Reading file contents

cat : displays the content of a file

head : displays the beginning of a file, by default 10 lines

tail : displays the end of a file, 10 lines default

less : returns content of a file one page at a time

space bar : move next page

b : move back one page

down arrow : move forward one line

up arrow : move back one line

q : quit

grep : searches a specified file and returns all lines in the file containing a specified string

| (piping) : send the standard output of one command as standard input to another command for further processing.

find : searches for directories and files that meet specified criteria.

find /home/analyst/projects -name "*log*" : case sensitive

find /home/analyst/projects -iname "*log*" : not case sensitive

find /home/analyst/projects -mtime -3 : based on days

* : used as a wildcard to represent zero or more unknown characters

mkdir : creates a dir

rmdir : removes a dir

touch : creates a file

rm : remove a file

mv : moves a file or directory to a new location

cp : copies a file or dir to a new location

nano : to edit files

Authorization: process of granting access to specific resources in a system

Permissions: the type of access granted for a file or a dir

Read = r

Write = w

Execute = x

Types of owners:

user = u

group = g

other = o

File permission are represented with a 10 character string

drwxrwxrwx - Directory with max permissions for each group

d : file type (dir)

user : read write execute

group : read write execute

other : read write execute

Options: modify the behavior of the command

ls -l : displays permissions to files and dir

ls -a : displays hidden files

ls -la : displays permissions to files and directories, including hidden files

change permissions

chmod : changes permissions on files and dir

chmod g+w, o-r access.txt

g : group

+ : add permissions

w : write

o : other

- : take away permissions

r : read

chown : changes owner ship of file or dir

sudo chown fgarcia access.txt

Root user (superuser): a user with elevated privileges to modify the system.

sudo: temporarily grants elevated permissions to specific users.

useradd : adds a user to the system

-g : gets the users default group, also called their primary group

sudo useradd -g security fgarcia

-G : adds the user to addition groups, also called supplemental or secondary groups

sudo useradd -G finance,admin fgarcia

userdel : deletes a user from the system

doesn't delete home dir

-r : delete user and home dir

usermod : modifies existing users

-d : changes users home dir

-l : changes users login name

-L : locks the account so user can't login

man : displays information on other commands and how they work

whatis : displays a description of a command on a single line

apropos : searches the manual page description for a specified string

Module 4

Database: an organized collection of information or data

- Accessed by multiple people simultaneously

- store massive amounts of data

- perform complex tasks while accessing data

spreadsheets

- designed for a single user or a small team

- store less data

Relational database: a structured database containing tables that are related to each other

primary key: a column where every row has a unique entry

foreign key: a column in a table that is a primary key in another table

Structured Query Language (SQL): A programming lang used to create, interact with, and request information from a database.

Query: a request for data from a database table or a combination of tables

Log: record of events that occur within an orgs systems

Syntax: the rules that determine what is correctly structured in a computing lang

SELECT : indicates which columns to return

FROM : which table to query

ORDER BY : sequences the records returned by a query based on a specified column or columns. This can be in either ascending or descending order.

Operator: A symbol or keyword that represents an operation

WHERE : indicates the condition for a filter

LIKE : used with WHERE to search for a pattern in a column

Data types:

String: data consisting of an ordered sequence of characters

Numeric: data consisting of numbers

DONT NEED QUOTATIONS

Date and time: data representing a date and/or time

BETWEEN : an operator that filters for numbers or dates within a range

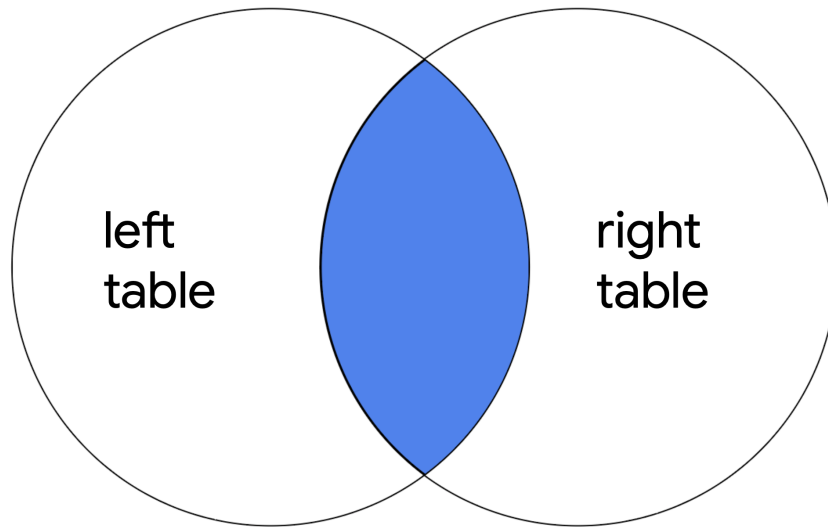
BETWEEN '18:00' AND '21:00'

AND : specifies that BOTH conditions must be met simultaneously

OR : specifies that either condition can be met

NOT : negates a condition

INNER JOIN : returns rows matching on specified column that exists in more than one table

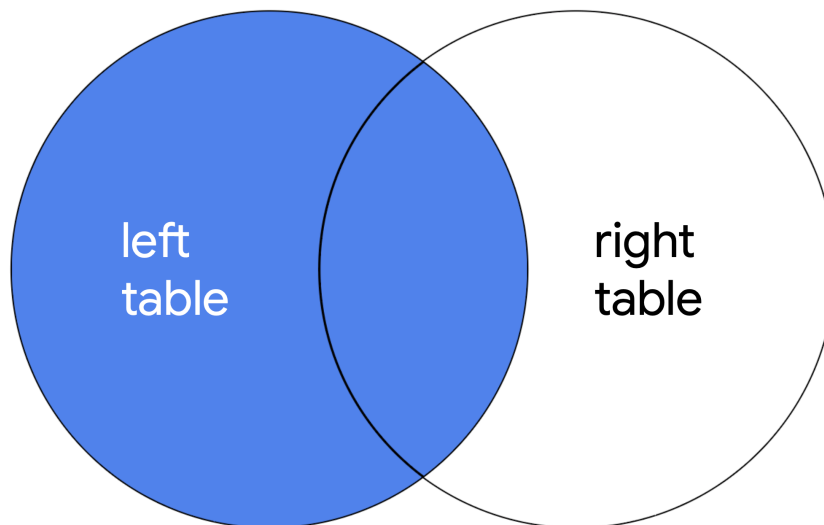


INNER JOIN machines ON employees.device_id = machines.device_id;

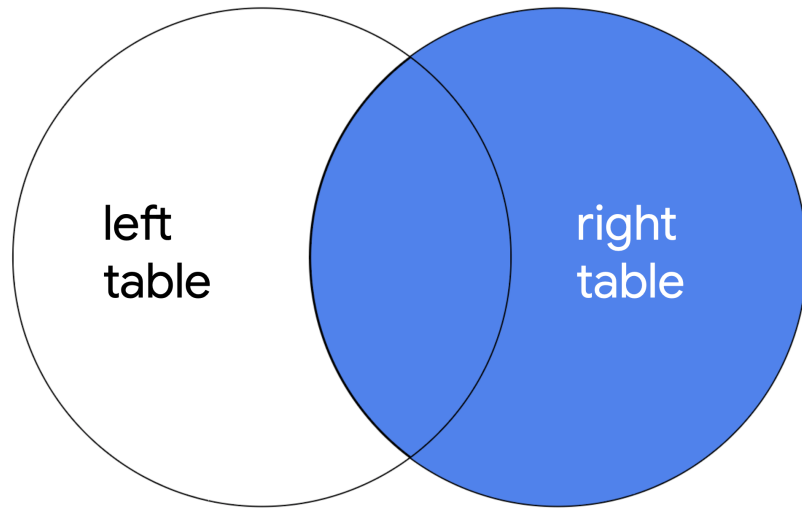
NULL represents a missing value for any reason

OUTER JOINS

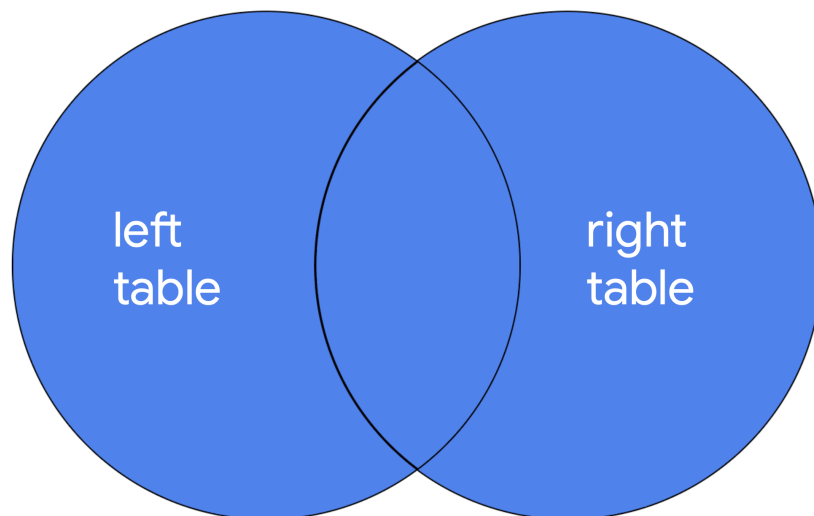
LEFT JOIN: returns all of the records of the first table, but only returns row of the second table that match on a specified column.



RIGHT JOIN: returns all of the records of the second table, but only returns rows from the first table that match on specified column



FULL OUTER JOIN: returns all records from both tables



Aggregate functions: functions that perform a calculation over multiple data points and return the result of the calculation. The actual data is not returned

COUNT : returns a single number that represents the number of rows returned from your query.

AVG : returns a single number that gives the avg of the numerical data in a column

SUM : returns a single number that gives the sum of the numerical data in a column